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RESEARCH ARTICLE

Impact of AI Literacy on Self-Efficacy Among Student Translators: Serial Mediating Roles of Self-Regulated Learning and Technology Acceptance

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ABSTRACT The recent advancement of artificial intelligence (AI) technology has revolutionized the language translation industry and unavoidably exerts an impact on student translators in multiple ways. In this context, this study investigated the antecedents of student translators' self-efficacy in AI-enhanced contexts, an underexplored area in current academic research. Data were collected via validated quantitative scales from 446 undergraduate student translators across eight public universities in China, who self-reported their AI literacy, technology acceptance, self-regulation, and self-efficacy in leveraging AI tools for translation tasks. Results indicate that AI literacy, technology acceptance, and self-regulation positively predicted translation self-efficacy. Furthermore, AI literacy served as a positive predictor of technology acceptance and self-regulation in AI-assisted translation, while technology acceptance positively predicted self-regulation. Additionally, technology acceptance and self-regulated learning (SRL) acted as partial mediators between AI literacy and translation self-efficacy, and the serial mediation pathway "AI literacy → technology acceptance → SRL → translation self-efficacy" was confirmed. This research offers theoretical and practical insights for translation education research and pedagogy.

INDEX TERMS AI literacy, technology acceptance, self-regulated learning, translation self-efficacy, student translators.

I. INTRODUCTION

Driven by AI technology, translator education has undergone a profound revolution [1]. Contemporary student translators are increasingly tasked with navigating a symbiotic workflow where AI tools function as dual-role supports: enhancing productivity through terminology mining and stylistic consistency verification during translation, while also serving as learning aids in post-editing translation outputs [2]. In response, some translation education programs have spearheaded the integration of AI-infused training to proactively equip students with the competencies required to meet the

evolving requirements of the translation industry [3]. However, this technological shift has concurrently introduced new challenges: while some students harness AI to surmount linguistic barriers and fortify task-specific confidence [4], others exhibit feelings of inadequacy regarding the usage of AI tools—a sentiment often amplified by over-reliance on such tools, exposure to inaccurate AI outputs, and encounters with ethical dilemmas when leveraging AI to support their learning [5]. This dichotomy underscores significant variability in students' translation self-efficacy [6], namely individuals' confidence in the ability to execute the requisite actions for producing an acceptable translation [7]. In this context, understanding the psychological mechanisms that shape student translators' self-efficacy in AI-augmented environments

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has become critical, as self-efficacy influences long-term career development in the evolving translation industry [8].

Despite the importance of academic self-efficacy, existing literature has predominantly zoomed in on traditional classroom settings, with limited attention to how AI technology reshapes its antecedent mechanisms. Prior studies have explored general factors that impact academic self-efficacy, such as teacher feedback [9], task complexity [10], and peer support [11], but the unique demands of AI-integrated translation—such as the need to critically evaluate AI outputs, independently adjust learning strategies, and adapt to evolving technological tools—call for a re-examination of the constructs that underpin self-efficacy in this new context. Specifically, student translators' ability to effectively engage with AI tools may depend on their mastery of domain-specific skills and attitudes, which interact dynamically to influence their confidence in producing high-quality translations [12]. Thus, this study examined the multifaceted antecedents of self-efficacy among student translators in AI-enhanced environments, aiming to inform targeted interventions that foster their psychological resilience and technological competence in translation practice.

II. THEORETICAL FOUNDATION: SOCIAL COGNITIVE THEORY

This research is grounded in Social Cognitive Theory (SCT), developed by seminal psychologist Albert Bandura. Central to SCT is the “triadic reciprocal determinism,” positing that personal cognitive processes (e.g., self-efficacy beliefs), environmental factors (e.g., physical resources and social feedback), and behavioral manifestations (e.g., action choices and execution outcomes) function not through unidirectional causality but as a dynamically interacting system [13], [14]. A critical mechanism bridging cognition and behavior within this framework is self-regulation: individuals actively manage their learning through iterative cycles of goal setting, process monitoring, strategy adjustment, and outcome reflection [15], [16]. By synthesizing cognitive processes, social interaction, and behavioral regulation, SCT has emerged as a canonical framework for explaining human learning, motivation, and behavioral change—with particular relevance to exploring the integration of educational technology into language instruction [17], [18].

Within SCT, self-efficacy emerges as the most central construct, defined as “beliefs in one's capabilities to organize and execute the courses of action required to manage prospective situations” [19]. Four experiential sources underlie the development of self-efficacy: direct mastery experiences (past successes), vicarious experiences (observing modeled behaviors' outcomes), verbal persuasion (social support/feedback), and physiological-affective states (reduced anxiety and positive emotional arousal) [20], [21]. In the era of AI, students' learning ecology has undergone profound transformations, with a growing number of learners integrating AI tools across diverse learning scenarios [22]. As social and environmental resources, AI tools can deliver real-time feedback and

present accurate exemplars to learners, thereby facilitating the acquisition of vicarious experience, mitigating learning anxiety, enabling continuous adjustment of learning rhythms, and ultimately fostering the attainment of mastery experiences [23], [24]. To date, academic attention has increasingly turned to probing learning self-efficacy within AI-supported educational contexts [24], [25], [26]. However, the bulk of research has been conducted within general learning or domain-general language learning contexts, with limited scholarly attention directed toward the domain-specific field of translation education. Despite this gap, we posit that findings concerning the antecedents of self-efficacy and the interplay among these constructs may be transferable to translation education settings. Consequently, the present study sought to extend the influencing mechanisms of academic self-efficacy to the AI-augmented translation context. In subsequent sections, we synthesize the relationships among these variables as documented in extant literature—spanning general learning research and language learning studies—to derive hypotheses for the current investigation.

III. LITERATURE REVIEW AND HYPOTHESES DEVELOPMENT

A. AI LITERACY AND LEARNING SELF-EFFICACY

Against the backdrop of exponentially expanding AI applications in educational contexts, scholars have turned their attention to learners' AI literacy. Conceptually, it refers to an individual's comprehensive competencies in critically evaluating and collaboratively deploying AI tools [27]. AI literacy encompasses four core dimensions: (1) awareness—the ability to identify and understand AI; (2) usage—the capacity to proficiently apply AI tools to complete tasks; (3) evaluation—the competence to analyze, select, and critically evaluate AI tools and their outcomes; and (4) ethics—the capability to recognize responsibilities and risks associated with AI technology use [28].

Learners' AI literacy is closely associated with their academic self-efficacy, which has been empirically evidenced. Illustrative of this, Zhang et al. found that AI literacy significantly predicted self-efficacy in AI-enhanced EFL learning, with AI learning self-efficacy serving as a full mediator between AI literacy and willingness to communicate among Chinese college EFL learners [24]. In a parallel investigation, Pan and Zhang explored factors influencing second language (L2) writing motivation among Chinese university students in AI-mediated writing environments. Their results indicated that AI literacy positively predicted writing self-efficacy, while writing self-efficacy partially mediated the association between AI literacy and writing motivation [29]. Collectively, these findings confirm that AI literacy functions as a robust positive predictor of language learners' self-efficacy; however, none of these investigations were conducted specifically within translation education. Zhang and Doherty investigated Chinese student translators' AI literacy and revealed that while students exhibited proficiency in AI translation knowledge and tool utilization—coupled

with confidence in identifying and rectifying AI-generated errors—they demonstrated limited awareness of AI's potential adverse impacts and associated ethical dilemmas [30]. Nevertheless, their study did not examine the relationship between student translators' AI literacy and translation self-efficacy. Drawing on previous literature, we propose:

H1: AI literacy is a positive predictor of translation self-efficacy in AI-enhanced environments.

B. SRL AND LEARNING SELF-EFFICACY

SRL represents a core construct underpinning diverse facets of the learning process [31]. It denotes learners' capacity to actively regulate cognition, motivation, and behavior toward learning goals through various activities [32]. Numerous empirical investigations have verified the influence of SRL on academic self-efficacy. For instance, Cho and Kim confirmed a positive association between self-efficacy beliefs and SRL strategy utilization among Korean high school EFL learners [25]. Xu et al. surveyed Chinese college EFL learners in smart classrooms, showing that SRL exerted a positive effect on self-efficacy. They further identified self-efficacy as a full mediator between SRL strategies and learning engagement [33]. Parallel evidence emerges from Almayez et al.'s study of Saudi college EFL learners using online language applications: their analysis not only revealed SRL as a positive predictor of EFL learning self-efficacy but also indicated that SRL mediated the link between motivation and self-efficacy [34]. With the integration of AI technology into education, SRL has expanded to encompass higher-order self-regulation, requiring learners to critically assess AI outputs and balance human-machine collaboration [35]. Such advanced regulatory capacity is particularly critical in high-skill domains such as translation. Synthesizing these empirical findings derived from different language learning contexts, the following hypothesis is advanced:

H2: SRL positively predicts translation self-efficacy within AI-enhanced learning environments.

C. TECHNOLOGY ACCEPTANCE AND LEARNING SELF-EFFICACY

Technology acceptance is another potential antecedent of learning self-efficacy. Davis et al.'s Technology Acceptance Model (TAM) [36] is acknowledged as a robust theory for interpreting users' acceptance of technological systems [37]. TAM comprises four key constructs: perceived usefulness (the extent to which users cognitively assess the benefits derived from using a technology), perceived ease of use (users' evaluation of the effort needed to operate the technology), attitude toward use (users' overall evaluative judgment of the technology), and behavioral intention (users' readiness for actual usage) [38]. Within TAM, perceived usefulness and ease of use collectively shape users' attitude toward a specific technology and their behavioral intention to adopt it; attitude toward use, in turn, exerts a direct influence on behavioral intention, which ultimately translates into actual technology usage [39].

Empirical research has examined the interplay between learners' technology acceptance and learning self-efficacy, yet the literature presents inconsistent findings. In general learning contexts, Le analyzed Vietnamese undergraduates' acceptance of ChatGPT and its impact on academic self-efficacy. The results indicated that students' attitudes toward ChatGPT directly predicted academic self-efficacy, whereas perceived ease of use and perceived usefulness exerted only indirect effects on academic self-efficacy via attitudes toward the technology [40]. Within EFL contexts, Huang et al. investigated the effects of Chinese university EFL learners' generative AI acceptance, perceived teacher enthusiasm, and self-efficacy on their EFL well-being. Their findings revealed that generative AI acceptance positively predicted learners' language learning self-efficacy across both productive and receptive skills [41]. In contrast, Yulian et al. found no significant effect of augmented reality technology acceptance on English learning self-efficacy among Indonesian university students enrolled in English for specific purposes courses [26]. Given these unresolved inconsistencies in the relationship between technology acceptance and self-efficacy, coupled with the lack of attention to the association between student translators' acceptance of AI technology and their translation self-efficacy, further empirical investigation is warranted. Building on this body of research, the present study proposes the following hypothesis:

H3: Student translators' acceptance of AI technology positively predicts their translation self-efficacy.

D. AI LITERACY AND TECHNOLOGY ACCEPTANCE

Beyond the intrinsic attributes of AI tools, the complexity of technology acceptance is fundamentally shaped by users' AI literacy—encompassing cognitive understanding, practical application, and critical evaluation of AI [42]. This relationship between AI literacy and technology acceptance has garnered empirical support in general educational contexts. Across diverse settings, studies converge on AI literacy's predictive power: Pan et al. demonstrated its impact on perceived ease of use, perceived usefulness, and behavioral intention among Chinese scholars using AI-assisted literature reading [43]; Gedik et al. identified AI literacy as explaining 17.5% of the variance in positive AI attitudes among individuals in Turkish universities [44]; and Salhab confirmed a direct effect of AI literacy on generative AI acceptance in a sample of Chinese undergraduates [45]. Though focusing on technology/information literacy rather than AI literacy, Mwakapesa's study of Chinese university students also aligns with this pattern, showing that technology literacy and information literacy directly predicted acceptance of ChatGPT [46]. Particularly germane to the present translation-focused research is Khoso et al.'s study of Chinese EFL learners. While their construct of "digital literacy" differs from AI literacy, their finding that digital literacy predicted perceived usefulness and ease of use offers a valuable parallel—suggesting that literacy in technology tools may similarly shape the acceptance behaviors of

language learners, including student translators [47]. Based on this empirical evidence, we hypothesize:

H4: Student translators' AI literacy positively predicts their acceptance of AI technology.

E. AI LITERACY, TECHNOLOGY ACCEPTANCE AND SRL

Within the educational landscape, empirical inquiries have corroborated the predictive power of AI literacy and technology acceptance for SRL. It is posited that AI literacy enhances learners' adaptive capacity in engaging with generative AI tools; through the active selection and strategic application of such technologies, learners can regulate and optimize their learning processes, thereby improving both the efficiency and quality of academic outcomes [48]. Empirically, Ji et al. indicated that AI literacy significantly predicted SRL and that SRL mediated the relationship between AI literacy and innovative behavior [49]. Another study by Xi and Yang explored the interplay between AI literacy, foreign language enjoyment, and online SRL among Chinese undergraduate learners, revealing that AI literacy exerted a positive influence on online SRL, with foreign language enjoyment serving as a significant mediator in this relationship [50]. Focusing on domain-specific SRL, Liu et al. analyzed the psychological mechanisms linking AI literacy in digital multimodal composing (DMC) to writing SRL among Chinese EFL learners. Their results demonstrated that DMC-specific AI literacy positively predicted writing SRL and that creative self-concept mediated the association between DMC AI literacy and writing SRL [51]. However, whether the relationship between AI literacy and SRL remains valid in AI-mediated translation contexts has yet to be empirically validated. Integrating these empirical insights, we propose:

H5: Student translators' AI literacy positively predicts translation SRL in AI-enhanced environments.

Beyond the positive impact of AI literacy, high technology acceptance also guides learners' SRL behaviors [52]. Overall, empirical research has consistently substantiated the predictive role of technology acceptance in SRL, albeit with contextual variations in effect sizes and mechanisms. In general learning settings, Chou et al. found that Chinese college students' technology acceptance positively predicted their self-regulation in augmented reality (AR)-supported learning environments [53]. Within EFL contexts, Pan revealed that Chinese EFL undergraduates' technology acceptance directly predicted attitudes toward technology-based self-directed learning, with learning motivation serving as a significant mediator in this relationship [54]. Another study by An et al. demonstrated that technology acceptance exerted a direct positive effect on SRL among Chinese middle school students, with intrinsic motivation and learning engagement dimensions (vigor, dedication, absorption) serving as parallel mediators [52]. However, Supriyono et al. reported mixed findings among Indonesian EFL student teachers in technology-mediated environments: perceived ease of use and internet self-efficacy positively predicted SRL, whereas perceived usefulness showed no significant

predictive power [55]. Notably, the specific relationship between technology acceptance and SRL in AI-mediated translation contexts remains unexplored. Thus, the present study hypothesizes:

H6: Student translators' technology acceptance positively predicts translation SRL in AI-enhanced environments.

IV. THE CURRENT RESEARCH

This study intended to empirically test the hypotheses derived from the literature review, centering on the antecedents of student translators' self-efficacy and the interplay of four core constructs—AI literacy, technology acceptance, SRL, and translation self-efficacy—in AI-augmented translation contexts. To date, no prior empirical investigation has simultaneously examined the dynamic relationships among these four variables, nor has their generalizability from general learning or language learning contexts to translation education been systematically validated. These gaps justify the necessity for the present study.

The research work is grounded in SCT, which emphasizes the principle of “triadic reciprocal determinism.” This framework assumes that personal cognitive factors (e.g., translation self-efficacy), environmental influences (e.g., AI tool utilization), and behavioral patterns (e.g., SRL strategies) function as an interdependent and dynamically interacting system [13]. By integrating SCT with the identified research gaps, this study sought to provide valuable theoretical and practical insights for AI-enhanced translation education. The model displaying the hypothetical relationships among the constructs is depicted in Figure 1.

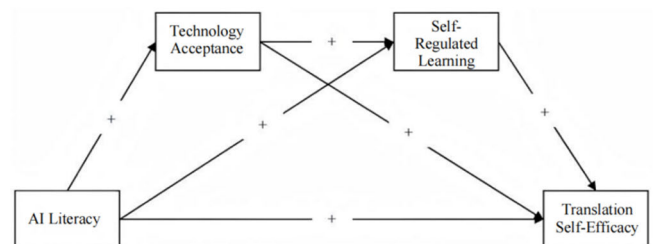


FIGURE 1. The hypothesized model.

The hypothesized model illustrates that, in addition to the direct predictive relationships among constructs, technology acceptance and SRL may link AI literacy to translation self-efficacy through both parallel and chained mediation pathways.

V. METHODOLOGY

A. PARTICIPANTS AND PROCEDURE

This study employed a quantitative approach to examine the antecedents of student translators' self-efficacy in AI-enhanced contexts. Data were gleaned through administering an online questionnaire survey, with participants recruited through a hybrid sampling strategy combining convenience sampling and teacher referrals. The inclusion criteria for the

target population are specified as follows: (1) enrollment in a bachelor's program in translation studies at a university in China's mainland; (2) previous experience with using at least one AI tool (e.g., DeepSeek, ChatGPT, or Doubao) for translation purposes; and (3) voluntary participation with signed informed consent. The final sample comprised 446 students from eight universities across China, with demographic characteristics distributed as follows: (1) Regional and institutional diversity: two universities in North China, three in Central China, two in Southwest China, and one in South China; (2) Gender distribution: 87 males (19.51%) and 359 females (80.49%); (3) Age range: 17–25 years (Mean = 20.47, $SD=1.33$); (4) Academic standing: 55 freshmen (12.33%), 155 sophomores (34.75%), 185 juniors (41.48%), and 51 seniors (11.44%).

Data collection was conducted from mid-October to early November 2025. The researchers contacted translation faculty members at each university via WeChat or email and requested their assistance in distributing the questionnaire. The questionnaire content was uploaded to Wenjuanxing (a professional online survey platform), with the first page displaying an informed consent form that explicitly stated the academic purpose of data use, guarantees of anonymity, and the removal of personally identifiable information. Collaborating teachers shared the questionnaire link via course-specific WeChat groups, and a two-week data collection period was implemented, with one reminder sent to improve response rates.

B. INSTRUMENTS

A composite online questionnaire was used for data collection, consisting of six sections: (1) an informed consent form; (2) demographic information, including gender, age, university affiliation, and types of AI tools used for translation tasks; (3) AI literacy scale; (4) technology acceptance scale; (5) SRL scale; and (6) translation self-efficacy scale. All items measuring the four constructs were rated on a 5-point Likert scale (1 = “totally disagree,” 5 = “totally agree”).

1) AI LITERACY MEASURES

The Artificial Intelligence Literacy Scale for Chinese College Students (AILS-CCS), developed by Ma and Chen [27], was used to assess participants' AI literacy. This scale comprises four dimensions: awareness (4 items; e.g., “I understand the definition of artificial intelligence”); usage (3 items; e.g., “I am proficient in using AI tools”); evaluation (4 items; e.g., “I can evaluate the limitations of different AI tools”); ethics (4 items; e.g., “I always adhere to ethical principles when using AI tools”).

2) TECHNOLOGY ACCEPTANCE MEASURES

Participants' acceptance of AI technology in translation contexts was measured based on Davis et al.'s [36] and Teo's [56] scales regarding perceived usefulness and ease of use. To adapt to the context of this study, the original items were modified accordingly: “computers/system” and

“work/job” were replaced with “AI tools” and “translation work,” respectively, and contextual clarifiers (“when I am doing translation work”) were added where necessary. The adapted AI Technology Acceptance Scale—Translation contains 8 items (4 items per dimension). Sample items for each dimension are as follows: perceived usefulness (“Using [AI tools] will improve my [translation] work”); perceived ease of use (“I find it easy to get [AI tools] to do what I want them to do [when I am doing translation work]”).

3) MEASURES FOR SRL IN AI-ENHANCED TRANSLATION

Student translators' SRL was evaluated using an adapted version of Xia et al.'s four-item scale [57], originally developed for chatbot-assisted learning contexts. The original scale was modified by replacing “learning English” and “chatbot” with “doing translation work” and “AI tools,” respectively. A sample item is: “When [doing translation work] with [AI tools], I will normally try to identify the knowledge that I do not understand well.”

4) AI-ENHANCED TRANSLATION SELF-EFFICACY MEASURES

An adapted version of the Translating Self-Efficacy Scale (TSE), originally developed by Yang et al. [58], was used to measure participants' self-efficacy in AI-assisted translation. This 15-item scale comprises three factors: Efficacy in Internal Competence (EIC, 9 items), Psycho-physiological Components (PPC, 3 items), and Efficacy in External Competence (EEC, 3 items). The contextual modifier “When doing translation work with AI tools” was added to the beginning of each section of the original scale. Sample items for each factor are as follows: EIC (“I can identify subtle word meaning differences”); PPC (“I can work hard to become a competent translator”); and EEC (“I can overcome difficulties in translation technology use”).

To mitigate respondents' cognitive load, one expert linguist was invited to translate the English scales into Mandarin Chinese—the mother tongue of all participating students. Another linguist then conducted back translation to verify accuracy. Subsequently, an educational psychologist was requested to judge the face validity of the translated scales. Following expert review, a pilot study was administered to 37 student translators (8 males and 29 females), who were presented with the Chinese versions of the four scales and asked to evaluate item clarity and understandability. Based on synthesized feedback from experts and pilot participants, corresponding revisions were made to the scales.

C. DATA ANALYSIS

Upon completion of data collection, SPSS 27.0 and Mplus 8.11 were employed for data analysis. The process began with verifying data completeness by examining responses for missing values or invalid entries. Next, reliability assessments and confirmatory factor analysis (CFA) were carried out to evaluate the internal consistency, discriminant validity, and convergent validity of the scales employed to measure the target constructs. Following this, descriptive analysis,

TABLE 1. Descriptive statistics.

Variables	Possible Range	Min	Max	Mean	Midpoint	SD	Skewness	Kurtosis
AI literacy (AIL)	15-75	15	75	55.596	45	7.884	1.078	-0.092
Technology acceptance (TA)	8-40	11	40	29.529	24	4.777	0.080	0.172
Self-regulated learning (SRL)	4-20	8	20	15.476	12	2.464	-0.289	0.048
Translation self-efficacy (SE)	15-75	26	75	53.737	45	8.626	0.294	0.157

TABLE 2. Fit indices and reliability coefficients.

Scales	χ^2/df	CFI	TLI	SRMR	RMSEA	Cronbach's α
Cutoff point	< 3	> 0.90	> 0.90	< 0.08	< 0.08	> 0.70
AI literacy	1.980	0.909	0.913	0.071	0.064	0.902
Technology acceptance	1.386	0.969	0.938	0.065	0.072	0.938
Self-regulated learning	1.893	0.995	0.995	0.008	0.047	0.914
Translation self-efficacy	2.557	0.905	0.923	0.045	0.050	0.950

normality tests, and multicollinearity checks were performed. Finally, structural equation modeling (SEM) was applied to test the proposed hypotheses via the bias-corrected bootstrap approach with 5,000 resamples and a 95% confidence interval.

VI. RESULTS

A. RESULTS OF DATA CHECK

Preliminary dataset quality screening confirmed the completeness of all observations with no missing values. To enhance data reliability, Mahalanobis distance was utilized for outlier detection, leading to the identification and exclusion of five outlier observations (excluded from subsequent analyses). Given that all data were obtained from participants' self-reported questionnaires, Harman's single factor test was performed to diagnose potential common method bias. Results of unrotated principal component analysis showed the first common factor explained merely 36.4% of total variance, which fell below the 40% cutoff value [59]. This indicates the dataset was not significantly affected by common method bias.

B. RESULTS OF DESCRIPTIVE ANALYSIS

To examine the overall profiles of participants' AI literacy, AI tool acceptance, self-regulation, and translation self-efficacy in AI-mediated contexts, descriptive analyses were conducted. As shown in Table 1, all four constructs exhibited moderate levels, with their mean scores exceeding the respective scale midpoints. Additionally, the skewness and kurtosis values fell within the range of -2 to $+2$, suggesting the data followed a normal distribution and thus met the assumptions for subsequent parametric statistical analyses [60].

C. RELIABILITY AND VALIDITY OF THE MEASUREMENT TOOLS

To evaluate the construct validity and reliability of the measurement scales, CFA and reliability assessments were

performed. Table 2 demonstrates that all fit indices—including the chi-square to degrees of freedom ratio (χ^2/df), comparative fit index (CFI), Tucker-Lewis index (TLI), standardized root mean square residual (SRMR), and root mean square error of approximation (RMSEA)—met the recommended benchmarks, indicating good fit across all measurement models [60]. Moreover, Cronbach's alpha coefficients ranged from 0.902 to 0.950, all exceeding the 0.7 cutoff, further supporting the scales' internal consistency [61].

Three criteria were used to evaluate convergent validity: standardized factor loadings ($\lambda > 0.5$) [61], average variance extracted (AVE > 0.5) [62], and composite reliability (CR > 0.7) [63]. Table 3 shows that all λ values fell between 0.525 and 0.919 (exceeding the 0.50 threshold), CR values for all constructs ranged from 0.842 to 0.914 (surpassing the 0.70 benchmark), and AVE values varied from 0.618 to 0.728 (all meeting the 0.50 cutoff). These findings indicate that the measurement model had adequate convergent validity. For discriminant validity, the square root of AVE for each construct (0.786-0.853) was greater than the absolute correlation coefficients with other constructs (0.502-0.567, indicating moderate correlations in this study), thus satisfying Fornell and Larcker's criterion [62]. This substantiates discriminant validity across the constructs.

D. RESULTS OF MODEL TESTING

SEM was applied to examine the predictive relationships among the four latent variables. Prior to this analysis, multicollinearity among variables was examined. The variance inflation factor (VIF) values all remained below the 5.0 threshold [64], indicating that multicollinearity did not pose an issue in the model. Besides, the model exhibited satisfactory goodness-of-fit, with all key indices meeting established statistical thresholds: $\chi^2/df = 2.378$, CFI = 0.937, TLI = 0.921, SRMR = 0.036, and RMSEA = 0.069. As presented in Table 4 and Figure 2, significant pathways emerged: (1) AI literacy positively predicted translation

TABLE 3. Convergent and discriminant validity.

Variables	Convergent Validity				Discriminant Validity			
	λ	p	AVE	CR	1	2	3	4
1. AI literacy	0.525-0.825	<.001	0.635	0.857	0.797			
2. Technology acceptance	0.701-0.919	<.001	0.641	0.842	0.502**	0.801		
3. Self-regulated learning	0.837-0.884	<.001	0.728	0.914	0.556**	0.567**	0.853	
4. Translation self-efficacy	0.718-0.865	<.001	0.618	0.869	0.517**	0.519**	0.564**	0.786

Notes: Numbers in bold represent the square root of AVE for each construct; ** $p < 0.05$

self-efficacy ($\beta = 0.460, p < .001, 95\%$ C.I. [0.100, 0.161]), supporting H1; (2) SRL positively predicted translation self-efficacy ($\beta = 0.321, p < .001, 95\%$ C.I. [0.093, 0.145]), confirming H2; (3) technology acceptance positively predicted translation self-efficacy ($\beta = 0.158, p < .001, 95\%$ C.I. [0.107, 0.290]), verifying H3; (4) AI literacy positively predicted technology acceptance ($\beta = 0.502, p < .001, 95\%$ C.I. [0.858, 1.381]), endorsing H4; (5) AI literacy positively predicted SRL ($\beta = 0.370, p < .001, 95\%$ C.I. [0.427, 0.587]), supporting H5; and (6) technology acceptance positively predicted SRL ($\beta = 0.371, p < .001, 95\%$ C.I. [0.372, 0.511]), supporting H6. Collectively, these findings confirm the hypothesized model. Additionally, AI literacy, technology acceptance and SRL together accounted for 63.3% of the variance in translation self-efficacy ($R^2 = 0.633, F(5, 435) = 150.023, p < .001$).

Given the confirmation of all direct pathways, the model incorporated two parallel mediation pathways and one serial mediation pathway. The results of mediation analysis in Table 5 reveal that the 95% confidence intervals (C.I.s) for the two parallel pathways—“AIL $\Rightarrow$ TA $\Rightarrow$ SE” (95% C.I. [0.038, 0.132]) and “AIL $\Rightarrow$ SRL $\Rightarrow$ SE” (95% C.I. [0.078, 0.180])—did not straddle zero, thereby verifying the presence of mediation effects [65]. Since AI literacy significantly predicted translation self-efficacy, technology acceptance (effect = 0.088, $z = 3.703, p < .001$) and self-regulated learning (effect = 0.133, $z = 5.251, p < .001$) emerged as two partial mediators between student translators’ AI literacy and self-efficacy. Furthermore, the pathway “AIL $\Rightarrow$ TA $\Rightarrow$ SRL $\Rightarrow$ SE” exhibited a significant partial chain mediation effect (effect = 0.064, $z = 4.609, 95\%$ C.I. [0.034, 0.088], $p < .001$). This result verifies that technology acceptance and SRL serially mediated the association between AI literacy and translation self-efficacy within AI-supported environments.

VII. DISCUSSION

The present study investigated the antecedents of translation self-efficacy among Chinese undergraduate student translators in AI-enhanced learning environments, with a specific focus on three key constructs: AI literacy, technology acceptance, and SRL. Additionally, it examined the dynamic interplay among these variables and systematically analyzed their direct and mediating effects on translation efficacy. In this section, the major findings are interpreted and discussed in light of the theoretical framework of SCT and relevant empirical literature.

The results show that participants’ AI literacy directly and positively predicted their self-efficacy in leveraging AI tools for translation tasks. As a bridge connecting individuals’ cognitive capacities with technological environments, this direct effect aligns with the proposition of SCT that environmental resources are converted into self-efficacy through individual capabilities [66]. Specifically, students with high AI literacy demonstrate deeper comprehension of and proficiency in utilizing AI tools to accomplish translation tasks. Through “mastery experiences” [20], they accumulate successful translation outcomes, thereby bolstering the efficacy belief: “I can efficiently complete translations using AI tools.” AI literacy serves as an essential element in modern education, equipping students to “navigate and succeed in a technology-driven world” [67]. Technological proficiency serves as a critical enabler in fostering students’ learning confidence during AI-integrated educational activities. By mastering relevant technical skills, learners gain enhanced agency in navigating AI-driven learning environments [24]. This autonomy, when combined with real-time personalized feedback generated by AI systems, strengthens learners’ perceived competence in utilizing such tools effectively, thereby enhancing their academic self-efficacy [24]. Importantly, AI literacy functions as a cognitive resource that extends beyond mere usage frequency. Its substantive depth—encompassing conceptual understanding, operational competence, and evaluative judgment—plays a decisive role in consolidating students’ self-belief regarding their capacity to successfully execute learning tasks [28]. In translation contexts, AI literacy manifests in the critical evaluation of AI outputs—for example, identifying grammatical inaccuracies or cultural misinterpretations in machine-translated texts. Such higher-order thinking skills solidify students’ confidence in managing translation quality, creating a positive feedback cycle that reinforces self-efficacy.

Additionally, technology acceptance directly and positively predicted translation self-efficacy. From a social cognitive perspective, acceptance constitutes a cognitive appraisal of environmental resources (e.g., AI tools). When students perceive AI tools as both useful—by shortening turnaround time and enhancing terminology accuracy—and easy to use—via intuitive interfaces and streamlined operations—they integrate these tools more actively into their translation workflows. This positive behavioral engagement, in turn, boosts self-efficacy beliefs [68]. Our findings replicate Huang et al.’s observation that AI acceptance predicts

TABLE 4. Direct effects.

X	→	Y	Estimate	SE	p	95% Confidence Interval		Results
						Lower Limit	Upper Limit	
AIL	→	SE	0.460	0.04	< .001	0.100	0.161	H1 Supported
SRL	→	SE	0.321	0.132	< .001	0.093	0.145	H2 Supported
TA	→	SE	0.158	0.046	< .001	0.107	0.290	H3 Supported
AIL	→	TA	0.502	0.035	< .001	0.858	1.381	H4 Supported
AIL	→	SRL	0.370	0.013	< .001	0.427	0.587	H5 Supported
TA	→	SRL	0.371	0.015	< .001	0.372	0.511	H6 Supported

Notes: → indicates a path influence relationship.

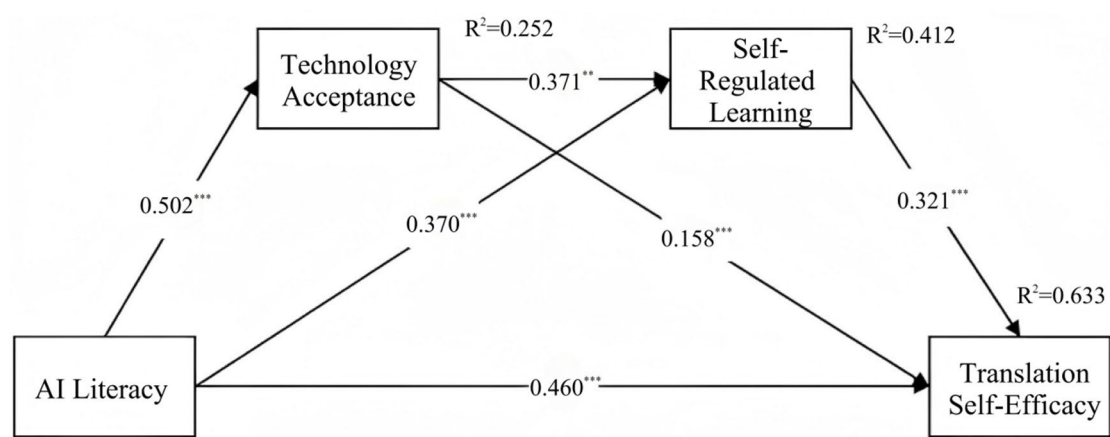


FIGURE 2. The tested model.
Notes: ***p<.001.

TABLE 5. Mediation effects.

Pathways	Effect	SE	95% Confidence Interval		z	p	Mediation
			Lower Limit	Upper Limit			
AIL→TA→SE	0.088	0.024	0.038	0.132	3.703	<.001	Partial
AIL→SRL→SE	0.133	0.025	0.078	0.180	5.251	<.001	Partial
AIL→TA→SRL→SE	0.064	0.014	0.034	0.088	4.609	<.001	Partial

self-efficacy in productive language skills among EFL learners [41]. Interactive functionalities of AI tools such as conversational agents and real-time error correction are particularly conducive to language production [24] due to their provision of immediate feedback and exemplars. These facilitate output refinement and the accumulation of vicarious and mastery experiences, which are known to enhance self-efficacy [69]. As a productive skill involving bilingual transfer and cultural adaptation, translation inherently entails multiple layers of complexity [70]; consequently, perceptions of usefulness and ease of use may translate particularly rapidly into positive self-efficacy appraisals. Notably, however, the present study found that technology acceptance ($\beta = 0.158$) exerted a significantly weaker influence on translation self-efficacy compared to AI literacy ($\beta = 0.460$) and SRL

($\beta = 0.321$). This may be attributed to the fact that, in translation contexts, students’ perceived usefulness and ease of use of AI tools play a less prominent role in boosting learning confidence. In contrast, the practical application of AI tools—such as frequency of use and strategic utilization—may more effectively enhance student translators’ self-efficacy when completing translation tasks with AI assistance. Based on SCT, although technology acceptance serves as a critical environmental assessment factor, the behavioral enactment (i.e., the frequency and strategic application of AI, which has a closer association with self-regulated learning, SRL) and deep-seated personal cognitive resources (AI literacy) act as more powerful and direct predictors of self-efficacy [13]. Nevertheless, this interpretation requires further empirical validation.

Our finding that SRL directly and positively predicted translation self-efficacy dovetails with the results of previous research [25], [33], which documented similar associations between SRL and language learning efficacy among EFL learners. Distinct from previous research, the present study is the first to extend this relationship to AI-mediated translation contexts. Collectively, these findings provide cross-contextual empirical evidence for Schunk and Zimmerman's argument that agentic learning experiences—generated through learners' proactive regulatory behaviors within the dynamic cycle of “goal setting–self-monitoring–strategic thinking” [71]—significantly enhance self-efficacy beliefs [72], [73]. This also aligns with the core tenet of triadic reciprocal determinism in SCT, which posits a dynamic and bidirectional interplay among individuals' cognitive processes and behaviors. Within the iterative cycle of SRL, individuals progressively accumulate mastery experiences—tangible achievements such as overcoming complex academic challenges or demonstrating measurable improvements in performance. These behavioral outcomes function as intrinsic positive reinforcement, directly bolstering individuals' perceived self-competence and serving as the foundational source of self-efficacy beliefs [13]. In AI-assisted environments, student translators may construct self-efficacy through the following SRL behaviors: (1) establishing phased human–machine collaboration goals; (2) systematically evaluating AI output across accuracy, fluency, and cultural appropriateness dimensions; (3) identifying stylistic patterns in AI-generated translations via comparative analysis (e.g., parallel text alignment); and (4) developing gap-specific revision strategies (e.g., prioritizing stylistic refinement for literary texts vs. terminology standardization for technical documents). This metacognitive monitoring process not only optimizes translation quality but, more critically, reinforces efficacy beliefs through mastery experiences [20]. Once again, these interpretations warrant confirmation through additional experimental studies.

The study further reveals that AI literacy positively predicted both technology acceptance and SRL, with technology acceptance also emerging as a significant positive predictor of SRL. AI literacy enhances students' cognitive understanding of AI tools (e.g., working principles and applicable scenarios), mitigates technological anxiety, and thereby elevates technology acceptance [74]. Concurrently, the critical evaluation fostered by AI literacy underpins the strategy selection and process monitoring phases of SRL [75]. The predictive effect of technology acceptance on SRL reflects an “environment-behavior” interaction in SCT [13]: upon recognizing AI tools' value and ease of use, students may proactively develop AI-assisted learning plans (e.g., leveraging AI for glossary expansion during translation) and monitor the effectiveness of tool usage, consequently enhancing SRL [52]. Collectively, these three direct predictive relationships lay the groundwork for subsequent mediation analysis regarding the pathways of “AI literacy → technology acceptance/SRL → translation self-efficacy.”

In terms of the parallel mediation, the “AI literacy → technology acceptance → translation self-efficacy” pathway underscores the bridging role of technology acceptance: AI literacy first strengthens students' perceptions of perceived usefulness and ease of use, transforming technology from an external tool into a trusted partner [76], and ultimately fortifying translation self-efficacy. Within translation contexts, students with adequate AI literacy (e.g., awareness of machine translation limitations) evaluate AI utility more rationally, avoiding blind dependence or outright rejection [77]. For example, high-AI-literacy students may adjust AI translation parameters (e.g., domain adaptation and style selection) to enhance output quality. This sense of control reinforces technology acceptance, translating into the efficacy belief that AI utilization improves translation competence. The “AI literacy → SRL → translation self-efficacy” pathway demonstrates AI literacy's empowering impact on SRL: AI literacy is not merely a technical skill but a metacognitive capability that enables students to better plan, monitor, and reflect on AI-assisted translation processes [78]. After generating draft translations via AI tools, student translators tend to compare AI output with reference texts to identify gaps in handling culture-specific terms and design targeted learning plans—a SRL process in which AI tools provide diversified strategic support, while AI literacy acts as the critical link between tool usage and strategy deployment. This SRL process may ultimately enhance translation self-efficacy [33].

Another key finding of the present study is the chained mediation effect among the four variables: “AI literacy → technology acceptance → SRL → translation self-efficacy.” This pathway unveils a cascading mechanism of self-efficacy formation in AI-empowered environments: AI literacy first enhances technology acceptance, creating favorable environmental conditions for SRL; driven by positive technological attitudes, students actively utilize AI tools for self-regulation (e.g., AI-based translation quality assessment), ultimately elevating translation self-efficacy through SRL practices. This process aligns closely with Bandura's triadic reciprocal model of SCT, wherein personal factors or beliefs (i.e., AI literacy and translation self-efficacy), environmental evaluation (i.e., technology acceptance), and behavioral regulation (i.e., self-regulation in translation) dynamically interact [13]. When student translators possess high AI literacy, they are likely to perceive AI tools as useful and easy to use, subsequently leveraging them for self-regulation by setting revision goals—thereby potentially enhancing their self-efficacy in translation through vicarious and mastery experiences.

VIII. IMPLICATIONS, LIMITATIONS, AND FUTURE DIRECTIONS

This study makes two primary theoretical contributions. First, it pioneers the integration of SCT's “person-environment-behavior” triadic interaction model with AI literacy, technology acceptance, SRL, and translation self-efficacy—

enriching the theory's applicability in digital contexts. Particularly, the identification of a serial mediation effect illuminates the complex mechanisms underlying self-efficacy formation in AI-enhanced environments, offering novel insights into the evolution of SCT within intelligent education paradigms. Second, it advances research on the antecedents of translation self-efficacy by shifting focus from traditional determinants (e.g., translation competence and teacher feedback) to AI-related variables. By systematically validating the pivotal roles of AI literacy, technology acceptance, and SRL, this study opens new avenues in translation educational psychology.

The study also carries practical implications for translation education. First, higher education institutions can implement tiered AI literacy curricula for student translators: foundational (AI tool operation), intermediate (AI output evaluation/revision), and advanced (AI ethics/cross-cultural adaptation). Case-based pedagogy—such as comparative analyses of AI and human translations—can foster technical awareness and critical thinking, laying groundwork for technology acceptance and SRL. Second, efforts can be made to improve the technological acceptance of AI tools. Educational administrators should select user-friendly AI translation platforms and provide targeted training (e.g., workshops on AI-assisted legal translation) to enhance perceived ease of use by lowering the barrier to entry. Simultaneously, empirical data demonstrating AI's efficiency gains can be leveraged to reinforce perceived usefulness, thereby facilitating the transition from technology acceptance to SRL. Third, instructors are recommended to cultivate students' AI-enabled SRL strategies through structured training in: (1) goal setting (e.g., AI drafting with manual refinement of cultural metaphors); (2) process monitoring (e.g., progress tracking via AI quality assessment tools); and (3) outcome reflection (e.g., maintaining AI usage logs to analyze tool strengths/weaknesses). Specifically, in concrete classroom activities, instructors could guide students to maintain a "human-AI collaboration journal" in which they document their entire translation process, including AI tool suggestions, their own revisions, and the underlying rationale for each modification. This practice has the potential to directly facilitate the outcome reflection phase of SRL.

Despite these implications, this study is not without limitations. First, the sample—drawn exclusively from eight Chinese universities—may introduce regional and cultural biases, while the cross-sectional design precludes insights into dynamic variable relationships (e.g., whether prolonged AI use alters SRL-self-efficacy associations). Second, reliance on self-report questionnaires raises concerns about social desirability bias. Third, the study did not differentiate between AI tool types (e.g., neural machine translation vs. computer-assisted translation tools) in their mediating effects. Finally, unexamined moderators (e.g., gender, translation experience, and AI usage frequency) may mask group-specific patterns (e.g., advanced students potentially relying more on SRL than technology acceptance).

To address these limitations, future research could: (1) adopt longitudinal mixed-methods designs (e.g., year-long cohort tracking with semi-structured interviews) to establish causality and deepen SRL practice insights; (2) refine constructs and tool typologies by developing a translation-specific AI literacy scale (e.g., adding the "cross-cultural AI translation evaluation" dimension) and comparing mediation effects across tools (e.g., ChatGPT vs. professional translation software); (3) incorporate moderators (e.g., AI usage frequency and learning motivation) and cross-national comparisons to enhance generalizability; and (4) design intervention experiments testing integrated AI literacy-SRL training programs via experimental-control group comparisons.

IX. CONCLUSION

The advent of AI technology turns out to be both a boon and bane for language teaching and learning, and it exerts profound impacts on the translation industry. For student translators, the optimal approach lies not in total avoidance of AI tools but in resourceful utilization to streamline translation tasks and enhance translation efficacy. Against this backdrop, this study conducted a preliminary inquiry into the antecedents of student translators' self-efficacy in AI-augmented contexts and tested a hypothesized model involving the interplay of AI literacy, technology acceptance, SRL, and translation self-efficacy. To enhance student translators' self-efficacy, higher education translation programs are recommended to cultivate their AI literacy, foster their technology acceptance, and strengthen their self-regulation capabilities in translation tasks. However, further research is warranted to validate the relationships among these variables, thereby providing evidence-based guidance for translation research and pedagogy.

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